Discrete Mathematics - Practical 4

Functions: worked solutions

with the reasoning spelled out, step by step

Covers the assigned subset of Practical 4: problems 2, 4, 5, 6, 10, 12, 19, 22, 24, 25.

Warm-up

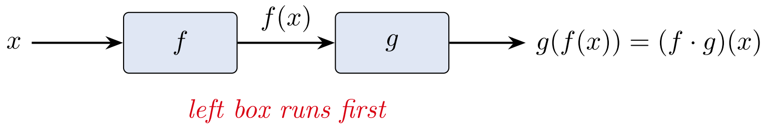
A *function* $f : A \to B$ is a rule that assigns to **each** input $x \in A$ exactly one output $f(x) \in B$. The set A is the **domain** (where the rule is allowed to act), and the set of values actually produced, $\{f(x) : x \in A\}$, is the **range** (or image). This sheet asks four kinds of question. (1) **Reading off domain and range** from a formula (problem 2): where is the formula legal, and what values come out. (2) **Composing** two functions and finding **inverses** (problems 4, 5). (3) **Classifying** a function as injective, surjective, or bijective (problems 6, 12). (4) **Reasoning about images and preimages** of sets (problems 10, 19, 22, 24, 25), where the whole game is whether f is injective.

One convention to fix before anything else: composition is left-to-right

In this course the dot product of functions is read **left-to-right**: $f \cdot g$ means "do f first, then g", so

$$(f \cdot g)(x) \;=\; g(f(x)).$$

This is the *opposite* of the symbol $\circ$ you may have seen, where $g \circ f$ means $g(f(x))$. We use the course convention everywhere below; problem 4 is where it bites, so watch the order there. Think of it as a pipeline: the input flows through the *left* box first.



Problem 2 - Domain and range from a formula

Problem 2

Let $f \subseteq \mathbb{R} \times \mathbb{R}$. Find the domain and the range for each formula:

$$\text{(a) } x^2 + 3, \quad \text{(b) } \sqrt{x-2}, \quad \text{(c) } \tfrac{1}{\sqrt{x-2}}, \quad \text{(d) } |x|, \quad \text{(e) } \tfrac{1}{x^2+2}, \quad \text{(f) } \tfrac{1}{x^2-2}.$$

The idea

Two separate questions, read off the graph. The **domain** is "where is the formula legal": avoid only two dangers, a *negative under a square root* and a *zero in a denominator*; everything else is allowed, so the domain is all of $\mathbb{R}$ minus those forbidden points. The **range** is "what heights does the graph reach": build it up from the inside out. For a formula like $\frac{1}{x^2-2}$, first see what $u = x^2 - 2$ can be, then see what $\frac{1}{u}$ does to that set. The reciprocal flips the order of an interval and sends values near 0 to $\pm\infty$, which is exactly why the range of (f) splits into two pieces.

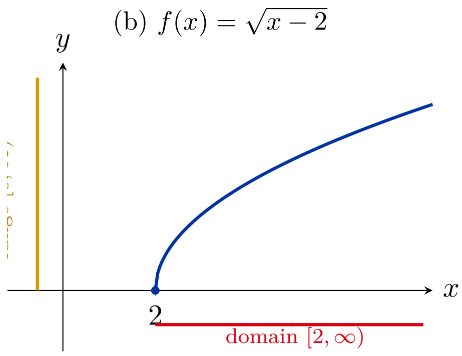
Step 1 - Domains (find the forbidden points). (a),(d),(e) have no roots and no denominators, and $x^2+2 \geq 2 > 0$, so they are legal everywhere: domain $\mathbb{R}$. For (b) $\sqrt{x-2}$ needs $x-2 \geq 0$, i.e. $x \geq 2$. For (c) $\frac{1}{\sqrt{x-2}}$ needs the root legal *and* nonzero, so $x-2>0$, i.e. $x>2$. For (f) $\frac{1}{x^2-2}$ needs $x^2-2 \neq 0$, i.e. $x \neq \pm\sqrt{2}$.

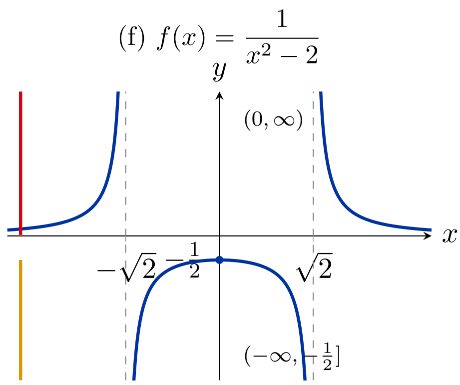
Step 2 - Ranges (work inside out). (a) $x^2 \geq 0 \Rightarrow x^2+3 \geq 3$, every value ≥ 3 is hit, range $[3,\infty)$. (b) $\sqrt{\cdot} \geq 0$ and grows without bound, range $[0,\infty)$. (c) $\frac{1}{\sqrt{x-2}} > 0$; as $x \to 2^+$ it blows up, as $x \to \infty$ it tends to 0 (never reaching it), range $(0,\infty)$. (d) $|x| \geq 0$, range $[0,\infty)$. (e) $x^2+2 \in [2,\infty)$, so $\frac{1}{x^2+2} \in (0,\frac{1}{2}]$; the maximum $\frac{1}{2}$ is at $x=0$, and the values shrink toward 0 as $|x| \to \infty$. Range $(0,\frac{1}{2}]$. (f) Let $u=x^2-2$, so $u \in [-2,\infty)\setminus\{0\}$. On $u \in [-2,0)$ the reciprocal gives $\frac{1}{u} \in (-\infty,-\frac{1}{2}]$ (the value $-\frac{1}{2}$ comes from $u=-2$, i.e. $x=0$); on $u \in (0,\infty)$ it gives $\frac{1}{u} \in (0,\infty)$. Range $(-\infty,-\frac{1}{2}] \cup (0,\infty)$.

Step 3 - Always check the boundary values. (e) at $x=0$: $\frac{1}{0+2}=\frac{1}{2}$. ✓ (f) at $x=0$: $\frac{1}{0-2}=-\frac{1}{2}$, the top of the negative branch. ✓ Numeric sampling on a fine grid confirms every min/max above.

Answer: (a) dom $=\mathbb{R}$, range $[3,\infty)$; (b) dom $=[2,\infty)$, range $[0,\infty)$; (c) dom $=(2,\infty)$, range $(0,\infty)$;
(d) dom $=\mathbb{R}$, range $[0,\infty)$; (e) dom $=\mathbb{R}$, range $(0,\frac{1}{2}]$; (f) dom $=\mathbb{R}\setminus\{\pm\sqrt{2}\}$, range $(-\infty,-\frac{1}{2}] \cup (0,\infty)$.

Reading domain and range off the picture. The two clearest cases, (b) and (f). For (b), the domain is the bracket on the x-axis ($x \geq 2$) and the range is the bracket on the y-axis ($y \geq 0$). For (f), the curve has vertical asymptotes at $x=\pm\sqrt{2}$ and splits the y-axis into exactly the two range pieces.





★ Method to remember

Transferable technique: domain = "all x except where a root goes negative or a denominator is zero." Range = build the value set *inside out*: track what the inner expression (x^2, $x-2$, x^2-2) can be, then apply the outer operation ($+3$, $\sqrt{\ }$, reciprocal). The reciprocal is the tricky one: it sends $(0,M]$ to $[\frac{1}{M},\infty)$ and a set straddling 0 to two separate pieces.

Problem 4 - Compositions $f \cdot g$ and $g \cdot f$

Problem 4

Let $f, g : \mathbb{R} \to \mathbb{R}$. Find $f \cdot g$ and $g \cdot f$ (course convention $(f \cdot g)(x) = g(f(x))$, apply f first):

$$\text{(a) } f(x) = x^2 + 1,\ g(x) = x + 3; \qquad \text{(b) } f(x) = \sqrt{x^2 + 2},\ g(x) = x^2 + 3;$$

$$\text{(c) } f(x) = \tfrac{1}{x},\ g(x) = 2x + 3.$$

The idea

Composition is just plugging one machine's output into the next machine. The only thing to get right is *which machine runs first*. Under the course's left-to-right rule, $f \cdot g$ runs f first: take x, compute $f(x)$, then feed that into g, giving $g(f(x))$. So to build $f \cdot g$, write down $g(\square)$ and replace every $\square$ by the whole formula for $f(x)$. Order matters: in general $f \cdot g \neq g \cdot f$, and part (a) shows that loudly.

Step 1 - Fix the recipe. $(f \cdot g)(x) = g(f(x))$ (substitute f into g); $(g \cdot f)(x) = f(g(x))$ (substitute g into f).

Step 2 - Part (a). $f(x) = x^2 + 1$, $g(x) = x + 3$.

$$(f \cdot g)(x) = g(f(x)) = \underbrace{(x^2 + 1)}_{f(x)} + 3 = x^2 + 4, \qquad (g \cdot f)(x) = f(g(x)) = \underbrace{(x + 3)}_{g(x)}{}^2 + 1 = x^2 + 6x + 10.$$

The two are different ($x^2 + 4 \neq x^2 + 6x + 10$), the headline lesson of this problem.

Step 3 - Part (b). $f(x) = \sqrt{x^2 + 2}$, $g(x) = x^2 + 3$.

$$(f \cdot g)(x) = g(f(x)) = \left(\sqrt{x^2 + 2}\right)^2 + 3 = (x^2 + 2) + 3 = x^2 + 5,$$

$$(g \cdot f)(x) = f(g(x)) = \sqrt{(x^2 + 3)^2 + 2}.$$

The square and the square root cancel cleanly in $f \cdot g$ because $x^2 + 2 \geq 0$.

Step 4 - Part (c). $f(x) = \frac{1}{x}$, $g(x) = 2x + 3$.

$$(f \cdot g)(x) = g(f(x)) = 2 \cdot \tfrac{1}{x} + 3 = \tfrac{2}{x} + 3, \qquad (g \cdot f)(x) = f(g(x)) = \tfrac{1}{2x+3}.$$

Step 5 - Always check at a sample point. (a) at $x = 1$: $f(1) = 2$, $g(2) = 5$, and $1^2 + 4 = 5$. ✓ For $g \cdot f$: $g(1) = 4$, $f(4) = 17$, and $1 + 6 + 10 = 17$. ✓ All three parts and both orders were confirmed numerically at many points.

Answer: (a) $f \cdot g = x^2 + 4$, $g \cdot f = x^2 + 6x + 10$; (b) $f \cdot g = x^2 + 5$, $g \cdot f = \sqrt{(x^2 + 3)^2 + 2}$; (c) $f \cdot g = \frac{2}{x} + 3$, $g \cdot f = \frac{1}{2x+3}$.

The arrow chain that makes the order unmistakable. For (a) at the sample $x = 1$, the two compositions trace different paths. Top row is $f \cdot g$ (the course order, f first); bottom row is $g \cdot f$.

$$f \cdot g\, x = 1 \xrightarrow{\text{apply } f} f(1) = 2 \xrightarrow{\text{apply } g} g(2) = 5 \qquad = (f \cdot g)(1)$$

$$g \cdot f\, x = 1 \xrightarrow{\text{apply } g} g(1) = 4 \xrightarrow{\text{apply } f} f(4) = 17 \qquad = (g \cdot f)(1)$$

Same start $x = 1$, different finishes (5 vs 17): composition is not commutative, and the left-to-right rule tells you which box fires first.

★ Method to remember

Transferable technique: to build $f \cdot g$ (course order), write the *second* function $g(\square)$ and substitute the *whole* first formula for $\square$. Always sanity check at one number by literally walking the pipeline. If your two compositions come out equal for a generic pair of functions, you almost certainly mixed up the order.

Problem 5 - Inverse functions

Problem 5

Find the inverse function for each:

$$\text{(a) } f(x) = \frac{x+4}{2}, \qquad \text{(b) } f(x) = x^3, \qquad \text{(c) } f(x) = \frac{x-2}{x+3}.$$

The idea

The inverse f^{-1} undoes f: if f sends x to y, then f^{-1} sends y back to x. To find a formula, just *solve* $y = f(x)$ *for* x. Whatever you get, $x =$ (expression in y), is $f^{-1}(y)$; rename y to x at the end if you like. Geometrically the inverse is the **mirror image of the graph across the line** $y = x$, because swapping the roles of input and output swaps the two axes.

Step 1 - Part (a). Solve $y = \frac{x+4}{2}$: multiply by 2, $2y = x+4$, so $x = 2y-4$. Hence $f^{-1}(x) = 2x-4$.

Step 2 - Part (b). Solve $y = x^3$: take the (real) cube root, $x = \sqrt[3]{y}$. Hence $f^{-1}(x) = \sqrt[3]{x}$. (Cube root is defined for all reals, including negatives, so this is a genuine inverse on all of $\mathbb{R}$ - which works precisely because $x^3 : \mathbb{R} \to \mathbb{R}$ is a bijection, as Problem 6(c) confirms.)

Step 3 - Part (c). Solve $y = \frac{x-2}{x+3}$. Clear the denominator: $y(x+3) = x-2 \Rightarrow yx + 3y = x - 2 \Rightarrow x(y-1) = -2-3y$, so

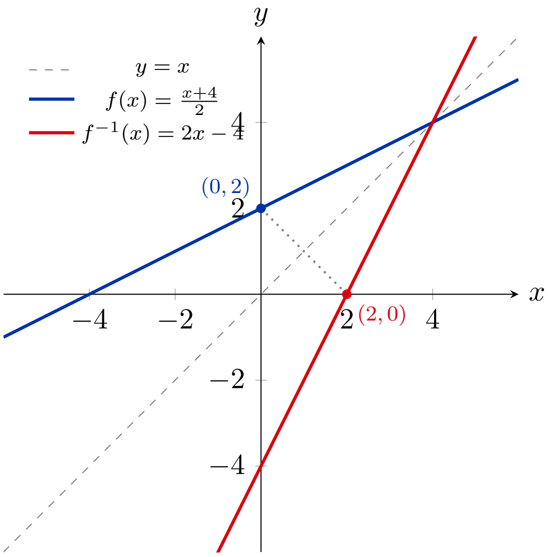
$$x = \frac{-2-3y}{y-1} = \frac{3y+2}{1-y}.$$

Hence $f^{-1}(x) = \dfrac{3x+2}{1-x}$ (defined for $x \neq 1$; the value 1 is exactly the height the graph never reaches, its horizontal asymptote).

Step 4 - Always check: f^{-1} really undoes f. (a) $f^{-1}(f(x)) = 2 \cdot \frac{x+4}{2} - 4 = (x+4) - 4 = x$. ✓ (c) take $x = 0$: $f(0) = \frac{-2}{3}$, then $f^{-1}(-\frac{2}{3}) = \frac{3(-2/3)+2}{1-(-2/3)} = \frac{0}{5/3} = 0$. ✓ Both directions verified numerically.

Answer: (a) $f^{-1}(x) = 2x-4$; (b) $f^{-1}(x) = \sqrt[3]{x}$; (c) $f^{-1}(x) = \dfrac{3x+2}{1-x}$ (for $x \neq 1$).

The inverse is a reflection across $y = x$. Here is part (a): the line $f(x) = \frac{x+4}{2}$ and its inverse $f^{-1}(x) = 2x-4$ are mirror images in the dashed diagonal. Pick any point on one graph, swap its coordinates, and you land on the other.



★ Method to remember

Transferable technique: to invert, *solve* $y = f(x)$ *for* x. For a fraction $\frac{ax+b}{cx+d}$, cross-multiply and collect all x terms on one side. Check by composing: $f^{-1}(f(x))$ must simplify to x. The inverse graph is always the reflection of f across $y = x$.

Problem 6 - Injective, surjective, bijective ($f : \mathbb{R} \to \mathbb{R}$)

Problem 6

Decide which of the maps $f : \mathbb{R} \to \mathbb{R}$ are injective, surjective, or bijective:

$$\text{(a) } |x|, \quad \text{(b) } x^2 + 4, \quad \text{(c) } x^3 + 6, \quad \text{(d) } x + |x|, \quad \text{(e) } x(x-2)(x+2).$$

The idea

Two independent questions, and each has a one-line picture test. **Injective** ("one-to-one") means *different inputs give different outputs*: no two x values share a height. Picture test: every **horizontal line** hits the graph *at most once*. If even one horizontal line cuts the graph twice, injectivity fails. **Surjective** ("onto $\mathbb{R}$") means *every* target height is reached: the range is all of $\mathbb{R}$. Picture test: every horizontal line hits the graph *at least once*. **Bijective** = both = every horizontal line hits *exactly once* (and then f has an inverse). The classic killers: a $|x|$ or even power folds the line in half (kills injectivity and surjectivity); an odd power like x^3 is a clean increasing curve (bijective).

Step 1 - The horizontal-line test, function by function.

(a) $|x|$: the line $y = 1$ meets it at $x = \pm 1$ (two points) $\Rightarrow$ *not* injective; the line $y = -1$ meets it nowhere ($|x| \geq 0$) $\Rightarrow$ *not* surjective. **Neither.**

(b) $x^2 + 4$: $f(1) = f(-1) = 5 \Rightarrow$ not injective; range is $[4, \infty)$, so $y = 0$ is never reached $\Rightarrow$ not surjective. **Neither.**

(c) $x^3 + 6$: strictly increasing (if $x_1 < x_2$ then $x_1^3 < x_2^3$), so every horizontal line meets it at most once $\Rightarrow$ injective; as $x \to \pm\infty$ the value runs over all of $\mathbb{R}$, so every line meets it at least once $\Rightarrow$ surjective. **Bijective**, with inverse $f^{-1}(x) = \sqrt[3]{x - 6}$.

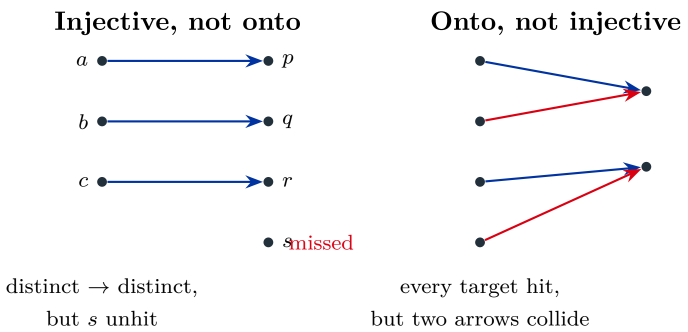
(d) $x + |x|$: for $x \le 0$ this is $x + (-x) = 0$, so the whole ray $x \le 0$ maps to $0 \Rightarrow$ badly non-injective; the output is never negative (it equals 0 or $2x > 0$) $\Rightarrow$ range $[0, \infty)$, not surjective. **Neither.**

(e) $x(x-2)(x+2) = x^3 - 4x$: it is a cubic, so as $x \to \pm\infty$ it covers all of $\mathbb{R} \Rightarrow$ surjective; but $f(0) = f(2) = f(-2) = 0$, so the line $y = 0$ meets it three times $\Rightarrow$ not injective. **Surjective only.**

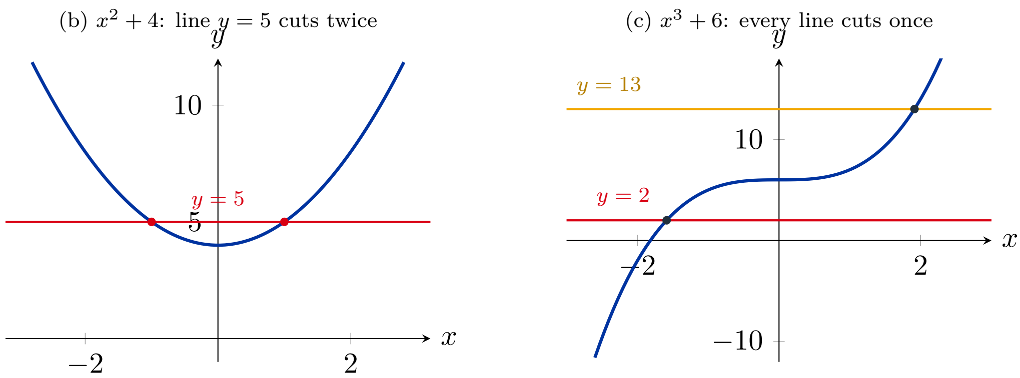
Step 2 - Always check the witnesses. The non-injective verdicts each rest on a concrete collision: $|-1| = |1|$, $(-1)^2 + 4 = 1^2 + 4$, $f(-2) = f(0)$ in (d), and $(-2)(-4)(0) = 0 = 2 \cdot 0 \cdot 4$ in (e). The non-surjective verdicts rest on a missed target ($y = -1$ for (a), $y = 0$ for (b)/(d)). For (c) the cube-root formula explicitly produces a preimage for every target. All confirmed by dense sampling.

Answer: (a) neither; (b) neither; **(c) bijective**; (d) neither; (e) surjective but not injective.

The showpiece: arrow diagrams and the horizontal-line test. First, the three pictures every student should carry in their head. Dots on the left are inputs, dots on the right are targets; an arrow is $x \mapsto f(x)$.



Now the horizontal-line test applied to the two contrasting graphs from this problem: $x^2 + 4$ (folded, a line can cut twice) versus $x^3 + 6$ (monotone, every line cuts exactly once).



★ Method to remember

Transferable technique: to disprove injectivity, exhibit one collision $f(a) = f(b)$ with $a \neq b$. To disprove surjectivity, exhibit one unreached target. To prove bijectivity of a real function, show it is strictly monotone (injective) and runs from $-\infty$ to $+\infty$ (surjective) - then it has an inverse. Even powers and $|x|$ fold; odd powers do not.

Problem 10 - $\mathrm{Ker}(f)$ is an equivalence relation

Problem 10

Show that for any function $f : A \to B$, its **kernel** $\mathrm{Ker}(f) = \{(x, y) \in A \times A : f(x) = f(y)\}$ is an equivalence relation on A.

The idea

The kernel relates two inputs exactly when they have the *same output*. But "having the same value" is built straight out of *equality* ($=$) of those values, and equality is the most equivalence-like relation there is: anything equals itself, equality is symmetric, and it chains. So $\mathrm{Ker}(f)$ inherits all three properties of $=$ for free. The payoff picture: $\mathrm{Ker}(f)$ chops the domain into **fibers**, one clump per output value, where a fiber $f^{-1}(b)$ is all inputs that f sends to b. Equivalence classes = fibers.

Step 1 - Reflexive. For any $x \in A$, $f(x) = f(x)$ trivially, so $(x, x) \in \mathrm{Ker}(f)$.

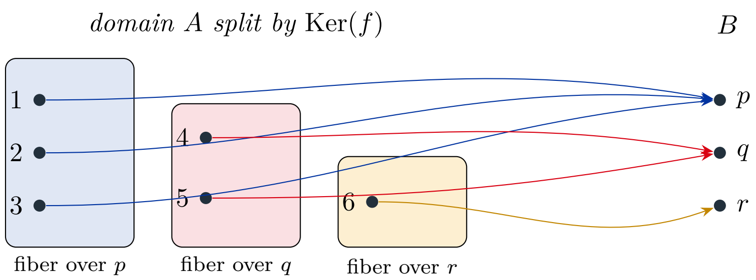
Step 2 - Symmetric. If $(x, y) \in \mathrm{Ker}(f)$ then $f(x) = f(y)$; equality is symmetric, so $f(y) = f(x)$, hence $(y, x) \in \mathrm{Ker}(f)$.

Step 3 - Transitive. If $(x, y), (y, z) \in \mathrm{Ker}(f)$ then $f(x) = f(y)$ and $f(y) = f(z)$; chaining equalities gives $f(x) = f(z)$, hence $(x, z) \in \mathrm{Ker}(f)$.

Step 4 - Conclusion and the fiber picture. All three hold, so $\mathrm{Ker}(f)$ is an equivalence relation. Its equivalence class of x is $\{y : f(y) = f(x)\} = f^{-1}(f(x))$, the fiber over the value $f(x)$. These fibers partition A: every input lands in exactly one fiber (its own output), and different output values give disjoint fibers.

Answer: $\mathrm{Ker}(f)$ is reflexive, symmetric, and transitive (inherited from $=$ on B), hence an equivalence relation on A; its classes are the fibers $f^{-1}(b)$.

The fiber partition. Take the toy map $f : \{1, 2, 3, 4, 5, 6\} \to \{p, q, r\}$ with $f(1) = f(2) = f(3) = p$, $f(4) = f(5) = q$, $f(6) = r$. The kernel partitions the domain into three fibers, one per value. Two inputs are $\mathrm{Ker}(f)$-related iff they sit in the same shaded clump.



★ Method to remember

Transferable technique: any relation of the form "$x \sim y$ iff $g(x) = g(y)$" ("same value of some quantity") is automatically an equivalence relation, because it is pulled back from $=$. Its classes are the level sets / fibers of g. This is the single most common way equivalence relations arise.

Problem 12 - $f(x) = \dfrac{x}{2x+1}$ on $\mathbb{Q}^+$ is injective but not surjective

Problem 12

Let $f : \mathbb{Q}^+ \to \mathbb{Q}^+$ be $f(x) = \dfrac{x}{2x+1}$. Show that f is injective but not surjective.

The idea

Injective is pure algebra: assume two inputs give the same output and grind the equation until it forces the inputs equal. **Not surjective** needs a target with no preimage, and the clean way to find one is to *bound the range*. For positive x, the denominator $2x + 1$ is a bit bigger than $2x$, so $\frac{x}{2x+1}$ is a bit smaller than $\frac{x}{2x} = \frac{1}{2}$. Every output is therefore strictly below $\frac{1}{2}$. So anything $\geq \frac{1}{2}$ (e.g. the value 1, which is in $\mathbb{Q}^+$) is unreachable - surjectivity fails by a wide margin.

Step 1 - Injective. Suppose $f(x) = f(y)$, i.e. $\dfrac{x}{2x+1} = \dfrac{y}{2y+1}$. Cross-multiply (both denominators are positive): $x(2y + 1) = y(2x + 1)$, so $2xy + x = 2xy + y$, hence $x = y$. Distinct inputs cannot share an output, so f is injective.

Step 2 - Bound the range. For $x \in \mathbb{Q}^+$ we have $x > 0$, so $2x + 1 > 2x > 0$ and therefore

$$f(x) = \frac{x}{2x+1} < \frac{x}{2x} = \frac{1}{2}.$$

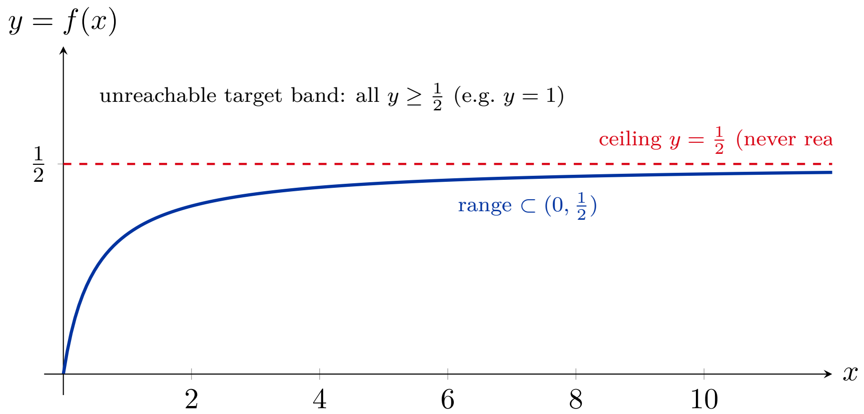
Also $f(x) > 0$. So the entire range sits inside the open interval $(0, \frac{1}{2})$.

Step 3 - Exhibit a missed target. The number $1 \in \mathbb{Q}^+$ satisfies $1 \geq \frac{1}{2}$, so it is outside $(0, \frac{1}{2})$ and cannot be a value of f. (Algebraically: $\frac{x}{2x+1} = 1$ forces $x = 2x+1$, i.e. $x = -1 \notin \mathbb{Q}^+$.) Hence f is *not* surjective.

Step 4 - Always check the bound is tight but never attained. As $x \to \infty$, $f(x) = \frac{x}{2x+1} \to \frac{1}{2}$ from below (e.g. $f(10000) \approx 0.49998$), so $\frac{1}{2}$ is the supremum but is never reached either - confirming the range is strictly inside $(0, \frac{1}{2})$.

Answer: Injective: $f(x) = f(y) \Rightarrow x = y$. Not surjective: every value is $< \frac{1}{2}$, so e.g. 1 (and any value $\geq \frac{1}{2}$) has no preimage.

The range sits below $y = \frac{1}{2}$. Plotting f for $x > 0$, the curve climbs from 0 but stays under the dashed ceiling $y = \frac{1}{2}$ forever. The whole target band $[\frac{1}{2}, \infty)$ of $\mathbb{Q}^+$ - including 1 - is empty of values.



★ Method to remember

Transferable technique: to show "not surjective," *bound the range* and name one target outside the bound. The estimate $\frac{x}{2x+1} < \frac{x}{2x}$ (shrink the denominator's $+1$) is the workhorse trick for rational functions. Injectivity for a ratio is usually a one-line cross-multiplication.

Problem 19 - When is $f(A \cap B) = f(A) \cap f(B)$?

Problem 19

For which functions f does $f(A \cap B) = f(A) \cap f(B)$ hold for all sets A, B?

The idea

One inclusion is free, the other costs injectivity. *Free direction:* if $x \in A \cap B$ then $f(x)$ is both in $f(A)$ and in $f(B)$, so $f(A \cap B) \subseteq f(A) \cap f(B)$ *always*. The reverse can fail because $f(A) \cap f(B)$ collects **shared output values**, which can come from *different* inputs - one input in A, a different one in B - that need not meet in $A \cap B$. Exactly when f is **injective**, a shared output forces a shared input, and the two sides become equal. So the identity holds for all A, B **iff** f **is injective**.

Step 1 - The always-true inclusion. Let $y \in f(A \cap B)$, so $y = f(x)$ with $x \in A \cap B$. Then $x \in A$ gives $y \in f(A)$, and $x \in B$ gives $y \in f(B)$, so $y \in f(A) \cap f(B)$. Hence $f(A \cap B) \subseteq f(A) \cap f(B)$ for *every* f.

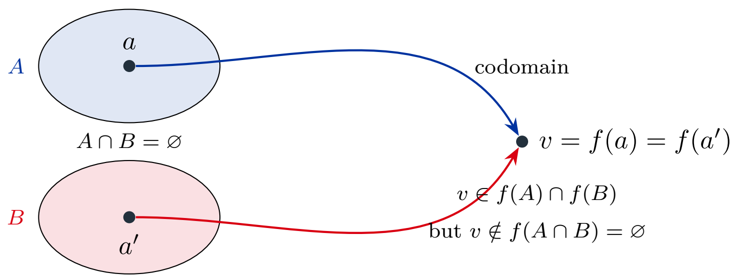
Step 2 - If f is injective, the reverse holds. Let $y \in f(A) \cap f(B)$, so $y = f(a) = f(b)$ with $a \in A$ and $b \in B$. Injectivity forces $a = b$, and this common point lies in $A \cap B$. Thus $y = f(a) \in f(A \cap B)$. So $f(A) \cap f(B) \subseteq f(A \cap B)$, and combined with Step 1 we get equality.

Step 3 - If f is not injective, the identity fails for some A, B. Pick $a \neq a'$ with $f(a) = f(a')$, and set $A = \{a\}$, $B = \{a'\}$. Then $A \cap B = \varnothing$, so $f(A \cap B) = \varnothing$, but $f(A) \cap f(B) = \{f(a)\} \cap \{f(a')\} = \{f(a)\} \neq \varnothing$. The identity breaks.

Step 4 - Always check on a concrete map. $f(x) = x^2$, $A = \{1\}$, $B = \{-1\}$: $f(A \cap B) = f(\varnothing) = \varnothing$ but $f(A) \cap f(B) = \{1\} \cap \{1\} = \{1\}$. The strict gap is visible, and it is exactly the non-injectivity ($1 \neq -1$ but $f(1) = f(-1)$) that opens it; for an injective f, Step 2 forces equality.

Answer: Exactly the **injective** functions. The inclusion $\subseteq$ always holds; equality holds for all A, B iff f is injective.

Why non-injectivity breaks it. Two different inputs $a \in A$ and $a' \in B$ collapse to the *same* output v. That output lands in both $f(A)$ and $f(B)$, so it sits in the intersection on the right - even though A and B share no input at all, leaving $f(A \cap B)$ empty.



★ Method to remember

Transferable technique: for image laws, the inclusion that needs no hypothesis is always "image of intersection $\subseteq$ intersection of images." The reverse needs injectivity, and the counterexample is always two collision points placed in different singletons. (Union behaves better: $f(A \cup B) = f(A) \cup f(B)$ for *every* f.)

Problem 22 - Can disjoint nonempty X_1, X_2 have $f(X_1) = f(X_2)$?

Problem 22

For $f : A \to B$ and disjoint nonempty $X_1, X_2 \subseteq A$ (so $X_1 \cap X_2 = \varnothing$), *can* the images be equal, $f(X_1) = f(X_2)$?

The idea

Images are about *output values*, not about which inputs produced them. Two completely separate input sets can still hit the very same set of outputs, provided f glues their points together. So as soon as f is non-injective, the answer is **yes, equal images are possible**. (Note the cleanest reading of the question: equal images are not *forced* - take $X_1 = \{1\}$, $X_2 = \{2\}$ under the identity to get unequal images - but they *can* happen, which is the interesting content.)

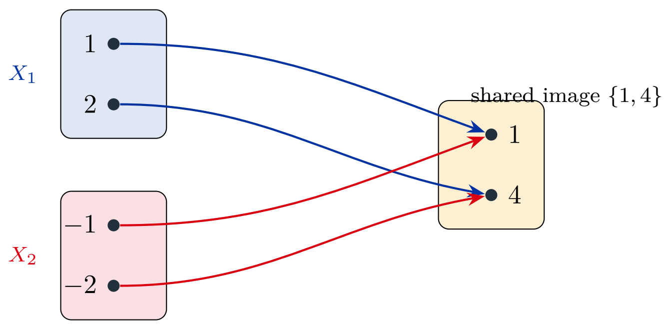
Step 1 - Build an example. Take $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$, and the disjoint singletons $X_1 = \{1\}$, $X_2 = \{-1\}$. They are nonempty and $X_1 \cap X_2 = \varnothing$.

Step 2 - Compute the images. $f(X_1) = \{1^2\} = \{1\}$ and $f(X_2) = \{(-1)^2\} = \{1\}$. So $f(X_1) = f(X_2) = \{1\}$, equal images from disjoint sets.

Step 3 - Always check the logic. The trick is exactly non-injectivity: f sends the distinct points 1 and -1 to the same value. With a larger example, $X_1 = \{1, 2\}$, $X_2 = \{-1, -2\}$ gives $f(X_1) = \{1, 4\} = f(X_2)$ too. (If f *were* injective, disjoint nonempty sets would always have unequal - indeed disjoint - images.)

Answer: **Yes**, it is possible. Example: $f(x) = x^2$, $X_1 = \{1\}$, $X_2 = \{-1\}$ are disjoint yet $f(X_1) = f(X_2) = \{1\}$. (Equal images are not forced, but they can occur whenever f is not injective.)

Picture: two disjoint sets, one shared image. The arrow diagram for $f(x) = x^2$ on $\{-2, -1, 1, 2\}$: the symmetric pair $\{1, 2\}$ and $\{-1, -2\}$ map onto the identical value set $\{1, 4\}$.



★ Method to remember

Transferable technique: "image of a set" forgets multiplicity and origin. Whenever a property is about images alone, a non-injective f (a folding map like x^2 or $|x|$) is your go-to source of surprises.

Problem 24 - Does $Y_1 \cap Y_2 = \varnothing$ force $f^{-1}(Y_1) \cap f^{-1}(Y_2) = \varnothing$?

Problem 24

Is it correct that for $f : A \to B$ and $Y_1, Y_2 \subseteq B$, if $Y_1 \cap Y_2 = \varnothing$ then $f^{-1}(Y_1) \cap f^{-1}(Y_2) = \varnothing$?

The idea

Preimages behave *perfectly* - much better than images - because each input x has exactly *one* output $f(x)$, so asking "is x in $f^{-1}(Y)$?" is just asking "is the single value $f(x)$ in Y?" If x were in both preimages, the one value $f(x)$ would have to live in both Y_1 and Y_2 at once - impossible when they are disjoint. No injectivity needed; this is **always** true. The deeper fact is $f^{-1}(Y_1) \cap f^{-1}(Y_2) = f^{-1}(Y_1 \cap Y_2)$ for every f.

Step 1 - Suppose the intersection is nonempty and derive a contradiction. Let $x \in f^{-1}(Y_1) \cap f^{-1}(Y_2)$. Then $f(x) \in Y_1$ and $f(x) \in Y_2$, so $f(x) \in Y_1 \cap Y_2$.

Step 2 - Use disjointness. But $Y_1 \cap Y_2 = \varnothing$, so $f(x)$ belongs to the empty set - impossible. Hence no such x exists, and $f^{-1}(Y_1) \cap f^{-1}(Y_2) = \varnothing$.

Step 3 - The clean reason (general identity). For any x:

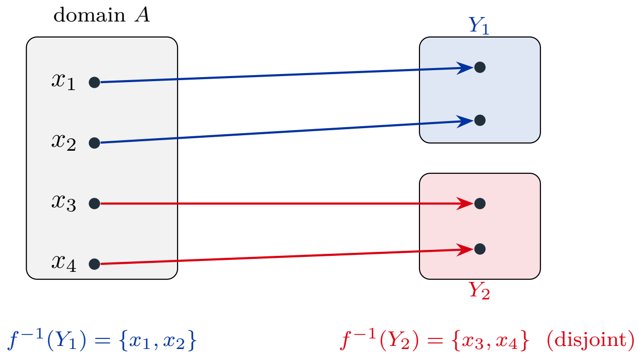
$$x \in f^{-1}(Y_1) \cap f^{-1}(Y_2) \iff f(x) \in Y_1 \text{ and } f(x) \in Y_2 \iff f(x) \in Y_1 \cap Y_2 \iff x \in f^{-1}(Y_1 \cap Y_2).$$

So $f^{-1}(Y_1) \cap f^{-1}(Y_2) = f^{-1}(Y_1 \cap Y_2)$ *always*, and the disjoint case is the special instance $Y_1 \cap Y_2 = \varnothing \Rightarrow f^{-1}(\varnothing) = \varnothing$.

Step 4 - Always check on a concrete map. Step 3 proves the identity for every f; a concrete instance with $f(x) = x^2$ ($Y_1 = \{1\}$, $Y_2 = \{4\}$ giving disjoint preimages $\{\pm 1\}, \{\pm 2\}$, whose union is $f^{-1}(\{1, 4\})$) bears out both the identity $f^{-1}(Y_1) \cap f^{-1}(Y_2) = f^{-1}(Y_1 \cap Y_2)$ and the disjoint consequence.

Answer: **Yes, always** (no hypothesis on f needed). Indeed $f^{-1}(Y_1) \cap f^{-1}(Y_2) = f^{-1}(Y_1 \cap Y_2)$, so disjoint Y's give disjoint preimages.

Why preimages cannot overlap here. Each input has a single output, so it can only sit in the preimage of whichever target block contains that one value. Disjoint target blocks $\Rightarrow$ each input is committed to at most one $\Rightarrow$ no input is shared.



> **★ Method to remember**
>
> **Transferable technique:** preimage commutes with *everything* - intersection, union, and complement - for *every* function: $f^{-1}(Y_1 \cap Y_2) = f^{-1}(Y_1) \cap f^{-1}(Y_2)$, similarly for $\cup$ and complement. This is because a preimage is defined by a single membership test on the one value $f(x)$. Images are the badly behaved direction; preimages are the friendly one.

Problem 25 - $f(f^{-1}(Y)) \subseteq Y$, with a strict example

> **Problem 25**
>
> Show that $f(f^{-1}(Y)) \subseteq Y$ for every $f : A \to B$ and $Y \subseteq B$. Give an example where the inclusion is strict.

> **The idea**
>
> Read the two operations literally. The preimage $f^{-1}(Y)$ is "all inputs whose output lands in Y." Apply f to that set and you get back outputs that, by construction, were already in Y. So you can only land back inside Y - never escape it. Why might you not fill all of Y? Because Y may contain target values that f *never produces* (values outside the range). Those have no inputs pointing at them, so $f(f^{-1}(Y))$ misses them, making the inclusion strict.

Step 1 - Prove the inclusion. Let $y \in f(f^{-1}(Y))$. Then $y = f(x)$ for some $x \in f^{-1}(Y)$. But $x \in f^{-1}(Y)$ means exactly that $f(x) \in Y$. So $y = f(x) \in Y$. As y was arbitrary, $f(f^{-1}(Y)) \subseteq Y$.

Step 2 - Build a strict example. Take $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$, and $Y = \{-1, 1\}$. Then

$$f^{-1}(Y) = \{x : x^2 \in \{-1, 1\}\} = \{x : x^2 = 1\} = \{-1, 1\}$$

(no real x has $x^2 = -1$), and applying f:

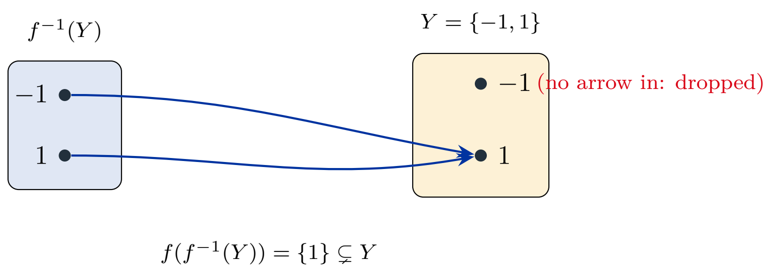
$$f(\{-1, 1\}) = \{1\}.$$

So $f(f^{-1}(Y)) = \{1\} \subsetneq \{-1, 1\} = Y$. The value $-1 \in Y$ is outside the range of f, so it is dropped.

Step 3 - Always check the boundary condition. Equality $f(f^{-1}(Y)) = Y$ holds exactly when $Y \subseteq f(A)$ (every element of Y is actually produced). Here $-1 \notin f(\mathbb{R}) = [0, \infty)$, so strictness is forced. Step 1 already proves $f(f^{-1}(Y)) \subseteq Y$ for every f; the example above is a concrete instance of the strict drop.

> **Answer:** $f(f^{-1}(Y)) \subseteq Y$ always. Strict example: $f(x) = x^2$, $Y = \{-1, 1\}$ gives $f(f^{-1}(Y)) = \{1\} \subsetneq Y$ (the value -1 is never an output).

Picture: the unreachable target gets dropped. Pull $Y = \{-1, 1\}$ back to its preimage, then push forward. Only 1 survives, because -1 has no arrow coming into it.



★ Method to remember

Transferable technique: the two round trips go opposite ways. Pushing then pulling *shrinks or stays*: $f(f^{-1}(Y)) \subseteq Y$ (equality iff $Y \subseteq f(A)$). Pulling then pushing *grows or stays*: $X \subseteq f^{-1}(f(X))$ (equality iff f is injective). Memorize the direction of each $\subseteq$ by asking "can a value/point be lost or gained?"